Data Storage Strategy Quiz

Multiple Choice Questions

For the streaming service's large, growing library of video files, movie stills, and captions, which storage type is the most scalable and cost-effective solution? Block Storage Object Storage A Relational Database File Storage

Answer

Object Storage. It excels at handling large, unstructured data (like videos and images) and is designed for high scalability and cost-efficiency.

2. The platform needs to perform high-performance tasks like video encoding and generating clips. Why is Block Storage the right choice for this processing step? 1. It is the cheapest option for long-term storage. 2. It is designed to store unstructured files with metadata. 3. It provides the high-performance, low-latency read/write operations needed for processing. 4. It is best for storing structured user data.

Answer

It provides the high-performance, low-latency read/write operations needed for processing.

3. How should structured user data, such as login credentials, subscription details, and viewing history, be stored? 1. In a database, which uses block storage as its underlying layer. 2. Directly in object storage as individual files. 3. On the same block storage used for video processing. 4. In flat files within the video library.

Answer

In a database, which uses block storage as its underlying layer.

4. Why do database systems (used for storing user data) typically rely on block storage? 1. Because block storage is the cheapest per-GB. 2. Because block storage is

ideal for large, unstructured media files. 3. Because databases need to frequently read and write small amounts of data with low latency. 4. Because object storage cannot be accessed over the internet.

Answer

Because databases need to frequently read and write small amounts of data with low latency.

True or False

5. True or False: After videos are processed (e.g., encoded into different qualities) using block storage, the final, processed video files should also be stored on block storage for long-term streaming. 1. True 2. False

Answer

False. The lesson states that long-term storage on block storage is too expensive, so the processed videos should be moved to object storage.

6. True or False: Object storage is the best choice for the high-performance processing tasks themselves, such as running video encoding scripts. 1. True 2. False

Answer

False. Block storage is ideal for high-performance processing. Object storage is ideal for storing the results of that processing.

Fill in the Blanks

7. _____ storage is the best solution for the large, unstructured, and ever-growing video library because it offers scalability, durability, and cost-efficiency. 1. Block 2. Object 3. Database 4. File

Answer

Object

8. High-performance, low-latency access makes _____ storage the right choice for the processing tasks, like video encoding. 1. Object 2. Relational 3. Block 4. File

Answer

Block

9. Structured user information, such as user IDs and viewing history, is best stored in a _____. 1. database 2. block file 3. object container 4. file system

Answer

database

10. In the proposed solution, a closed-caption text file would be stored as a separate _____ alongside its corresponding video file. 1. Block 2. Database entry 3. Object 4. Index

Answer

Object